

CODEALPHA

VIRTUAL INTERNSHIP PROJECT REPORT

AIRCRAFT WING DESIGN & CFD ANALYSIS

NACA 4412 AIRFOIL

Project Domain	Aerospace Engineering / Computational Aerodynamics
CAD Platform	SOLIDWORKS Premium 2025
CFD Platform	SOLIDWORKS Flow Simulation
Primary Deliverable	3D NACA 4412 Aircraft Wing + CFD Analysis
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Executive Summary

This project presents the design, three-dimensional CAD development, and computational fluid dynamics (CFD) analysis of an aircraft wing based on the NACA 4412 airfoil. The work was completed as part of a CodeAlpha Virtual Internship and was carried out using SOLIDWORKS Premium 2025 and SOLIDWORKS Flow Simulation.

The project follows an engineering workflow in which an established aerodynamic profile is translated into a three-dimensional wing geometry and subsequently evaluated computationally. The CFD post-processing supplied with the project shows a surface pressure contour together with flow trajectories around the wing. The simulation also reports a normal force of 574.656 N and a resultant force of 576.663 N.

The report combines the CAD/CFD work with the underlying mathematics. It explains the NACA four-digit airfoil formulation, conservation of mass, Bernoulli's relation, dynamic pressure, pressure-force integration, aerodynamic force coefficients, Reynolds number, and the governing Navier–Stokes equations. Importantly, only simulation quantities visible in the supplied project evidence are reported numerically; unavailable setup quantities are left symbolic rather than assumed.

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Nomenclature

Symbol	Definition
A	Area (m ²)
c	Airfoil chord length (m)
C _D	Drag coefficient
C _L	Lift coefficient
F	Aerodynamic force (N)
F _N	Normal-force component (N)
F _R	Resultant force (N)
g	Gravitational acceleration (m/s ²)
L	Lift (N)
D	Drag (N)
m	Maximum camber parameter
p	Location of maximum camber
P	Static pressure (Pa)
q	Dynamic pressure (Pa)
Re	Reynolds number
S	Reference wing area (m ²)
t	Maximum thickness parameter
V	Velocity (m/s)
μ	Dynamic viscosity (Pa·s)
ρ	Density (kg/m ³)
ν	Kinematic viscosity (m ² /s)

1. Introduction

Aircraft wings are aerodynamic lifting surfaces whose geometry strongly influences lift, drag, stability, and efficiency. The cross-sectional airfoil determines how the flow is accelerated, decelerated, and turned around the surface, while the three-dimensional planform controls the overall aerodynamic behavior.

Computer-aided design provides the geometric foundation for modern engineering analysis. Once a clean CAD model is available, CFD can be used to numerically predict flow behavior and surface quantities before physical testing. This project demonstrates that CAD-to-CFD workflow using a NACA 4412 wing.

The project is particularly relevant to aerospace engineering because it integrates geometry, fluid mechanics, aerodynamics, numerical simulation, and engineering interpretation rather than treating these subjects independently.

2. Internship and Project Context

The project was completed as part of a CodeAlpha Virtual Internship. The internship provided a practical setting for applying engineering concepts through a defined technical deliverable. The selected task focused on aircraft wing design and CFD analysis, allowing the theoretical study of airfoils and fluid mechanics to be connected with professional engineering software.

The final deliverable consists of a SOLIDWORKS aircraft wing model based on NACA 4412 together with CFD post-processing results. The work is documented in this report in a format intended to resemble a professional engineering project submission.

3. Project Definition

Project	Aircraft Wing Design and CFD Analysis
Airfoil	NACA 4412
CAD Software	SOLIDWORKS Premium 2025
CFD Software	SOLIDWORKS Flow Simulation
Analysis Type	External aerodynamic flow analysis
Primary Outputs	Pressure contour, flow trajectories, aerodynamic force
Normal Force	574.656 N
Resultant Force	576.663 N

Project Objectives

- Develop an accurate three-dimensional NACA 4412 aircraft wing in SOLIDWORKS.
- Prepare the wing geometry for external-flow CFD analysis.
- Use SOLIDWORKS Flow Simulation to obtain the flow solution and post-process the results.
- Visualize the surface pressure distribution and flow trajectories.
- Determine the aerodynamic force acting on the wing.
- Relate the numerical results to fundamental aerodynamic equations.
- Develop practical CAD, CFD, interpretation, and technical-reporting skills.

4. NACA 4412 Airfoil Theory

NACA 4412 belongs to the classical four-digit NACA airfoil family. In this notation, the first digit represents maximum camber as a percentage of chord, the second represents the chordwise location of maximum camber in tenths of chord, and the last two digits represent maximum thickness as a percentage of chord.

$$\text{NACA 4412: } m = 0.04, \quad p = 0.40, \quad t = 0.12$$

Thus, the airfoil has 4% maximum camber located at 40% of the chord and a maximum thickness of 12% of the chord.

Mean Camber Line

$$y_c = (m/p^2)(2px - x^2), \quad 0 \leq x \leq p$$

$$y_c = [m/(1-p)^2][(1-2p) + 2px - x^2], \quad p \leq x \leq 1$$

Thickness Distribution

$$y_t = 5t[0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4]$$

Here x is normalized chord position. The upper and lower surface coordinates are generated from the mean camber line and thickness distribution.

$$\theta = \tan^{-1}(dy_c/dx)$$

$$x_u = x - y_t \sin\theta, \quad y_u = y_c + y_t \cos\theta$$

$$x_l = x + y_t \sin\theta, \quad y_l = y_c - y_t \cos\theta$$

These equations provide the mathematical basis of the aerodynamic profile used to generate the wing geometry.

5. Mathematical and Aerodynamic Foundations

5.1 Conservation of Mass

$$\dot{m} = \rho AV$$

$$\text{For steady flow: } \rho_1 A_1 V_1 = \rho_2 A_2 V_2$$

For approximately incompressible flow, density is treated as constant and the relation reduces to $A_1 V_1 = A_2 V_2$. This explains why changes in flow area are accompanied by corresponding changes in velocity.

5.2 Bernoulli's Equation

$$P + \frac{1}{2}\rho V^2 + \rho gh = \text{constant}$$

When elevation changes are negligible, the relation becomes:

$$P + \frac{1}{2}\rho V^2 = \text{constant}$$

The equation provides an important first-order interpretation of the pressure field around an airfoil: under its assumptions, regions of higher local velocity correspond to lower static pressure.

5.3 Dynamic Pressure

$$q_\infty = \frac{1}{2}\rho V_\infty^2$$

Dynamic pressure provides the reference pressure scale used to non-dimensionalize aerodynamic forces.

5.4 Pressure Force on a Surface

$$d\vec{F} = -p \vec{n} dA$$

$$\vec{F}_p = -\int_S p \vec{n} dA$$

In a numerical CFD solution, the continuous surface integral is approximated over many small surface elements:

$$\vec{F}_p \approx -\sum (p_i \vec{n}_i \Delta A_i)$$

This is the physical basis for obtaining an integrated pressure force from the computed pressure field.

5.5 Lift, Drag, and Force Coefficients

$$L = \frac{1}{2} \rho V_\infty^2 S C_L$$

$$D = \frac{1}{2} \rho V_\infty^2 S C_D$$

$$C_L = L / (\frac{1}{2} \rho V_\infty^2 S), \quad C_D = D / (\frac{1}{2} \rho V_\infty^2 S)$$

The appropriate force components must be defined relative to the free-stream direction and the coordinate system used in the CFD project.

5.6 Reynolds Number

$$Re = \rho V_\infty c / \mu = V_\infty c / \nu$$

Reynolds number represents the relative importance of inertial and viscous effects and is one of the key parameters governing airfoil behavior.

5.7 Governing CFD Equations

At the continuum level, CFD is based on conservation laws. For incompressible flow, a compact form of the governing equations is:

$$\nabla \cdot \vec{V} = 0$$

$$\rho(\partial \vec{V} / \partial t + \vec{V} \cdot \nabla \vec{V}) = -\nabla p + \mu \nabla^2 \vec{V} + \rho \vec{g}$$

The software converts these continuous equations into a numerical problem over a computational mesh and solves for the flow field subject to the specified physical and boundary conditions.

6. CAD Modeling Methodology

The CAD stage converts the mathematical airfoil definition into a three-dimensional engineering model. The following workflow was used conceptually in the development of the wing:

- Create the SOLIDWORKS part environment and establish suitable reference geometry.
- Define the NACA 4412 airfoil profile.
- Use the profile as the cross-sectional geometry of the wing.
- Develop the three-dimensional wing surface/solid using appropriate SOLIDWORKS modeling features.
- Inspect the resulting geometry for continuity and suitability for CFD.
- Save the completed wing model as the basis for the Flow Simulation project.

A clean and watertight CAD geometry is essential for reliable external-flow CFD because unwanted gaps, self-intersections, or invalid surfaces can interfere with domain creation and meshing.

7. CFD Methodology

The CFD stage was performed using SOLIDWORKS Flow Simulation. The supplied project evidence shows a dedicated Flow Simulation project named “Aircraft Wing CFD analysis” with mesh, surface plots, flow trajectories, and surface-parameter results.

- Prepare the aircraft wing geometry for external flow.
- Define the computational region surrounding the wing.
- Specify the fluid and relevant flow conditions.
- Generate the computational mesh.
- Run the numerical solution.
- Create a surface pressure contour on the wing.
- Generate flow trajectories to visualize the surrounding flow.
- Evaluate surface parameters to obtain aerodynamic force quantities.

The analysis presented in this report is based on the supplied post-processing evidence. Quantities not visible in the evidence—such as a complete set of inlet conditions, reference area, or mesh statistics—are not assigned numerical values in this report.

8. CFD Results

The CFD post-processing provides the principal numerical and visual results of the project. The displayed pressure legend ranges from 100,553.29 Pa to 101,895.63 Pa. The surface-parameter result reports a normal force of 574.656 N and a resultant force of 576.663 N.

Quantity	Value	Unit
Minimum displayed pressure	100,553.29	Pa
Maximum displayed pressure	101,895.63	Pa
Pressure range	1,342.34	Pa
Normal force	574.656	N
Resultant force	576.663	N

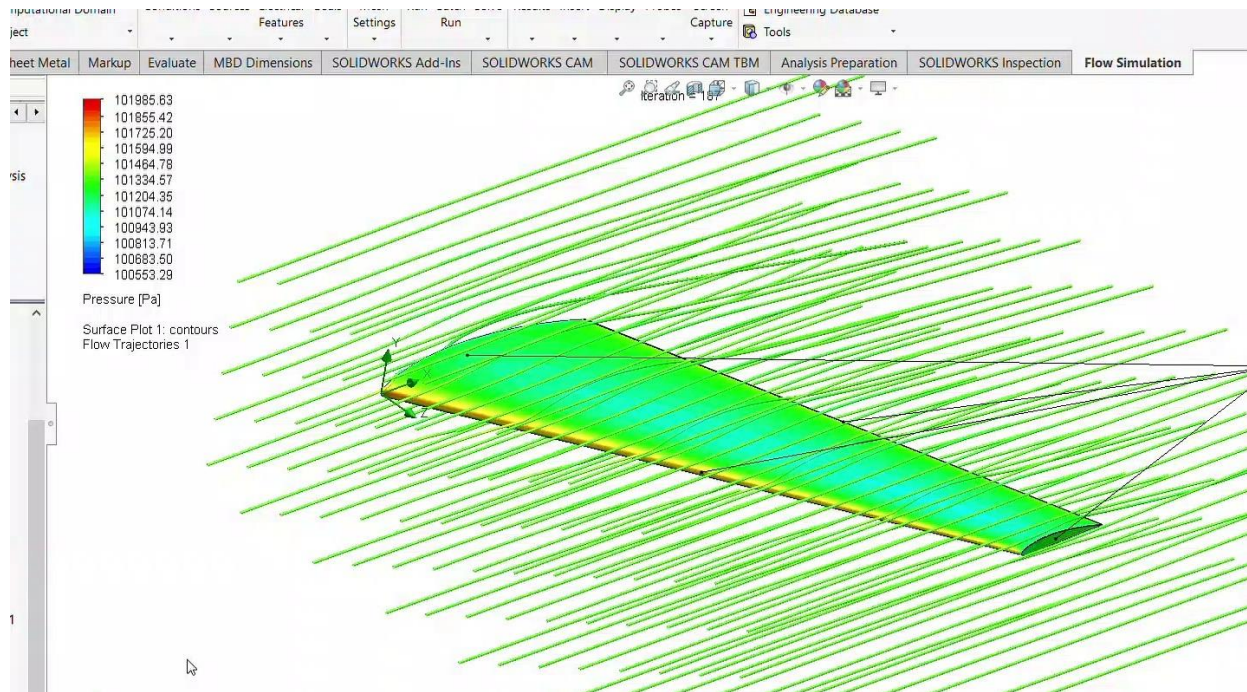


Figure 1. SOLIDWORKS Flow Simulation post-processing: pressure contour, flow trajectories, and reported force.

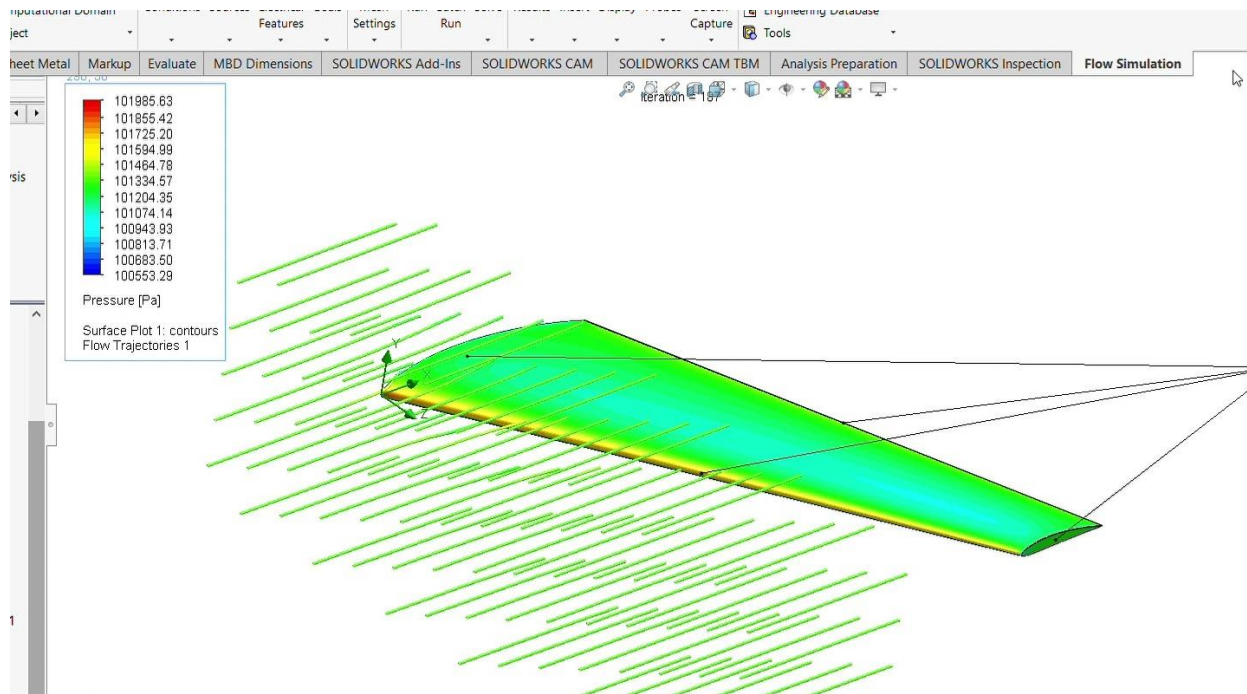


Figure 2. CFD visualization of the NACA 4412 wing surface and surrounding flow trajectories.

Interpretation of the Pressure Contour

The contour shows a spatially varying pressure field over the wing surface. The color variation indicates that the local pressure is not uniform. This pressure distribution is one of the main mechanisms responsible for the net aerodynamic force acting on the wing.

The flow trajectories provide qualitative information about the direction and organization of the flow around the three-dimensional wing. They are particularly useful for visually connecting the wing geometry with the computed flow field.

9. Quantitative Interpretation of Results

9.1 Pressure Difference Across the Displayed Range

$$\Delta P = P_{\text{max}} - P_{\text{min}}$$

$$\Delta P = 101,895.63 - 100,553.29 = 1,342.34 \text{ Pa}$$

This is the difference between the maximum and minimum pressure values displayed on the contour. It represents the range of the plotted pressure field; it should not be interpreted as a uniform upper-surface versus lower-surface pressure difference.

9.2 Force Comparison

$$F_R - F_N = 576.663 - 574.656 = 2.007 \text{ N}$$

$$(F_N/F_R) \times 100 \approx 99.65\%$$

The normal-force magnitude is approximately 99.65% of the displayed resultant-force magnitude. This indicates that, for the selected force definition and coordinate system, the resultant is dominated by the normal component.

The exact physical interpretation of “Normal Force” depends on the surface and coordinate definitions selected in the Flow Simulation project. It should therefore not automatically be labeled as lift unless the force direction is confirmed to coincide with the lift direction.

9.3 Relation Between Pressure and Aerodynamic Force

The CFD force is not obtained simply by multiplying the maximum pressure by the wing area. Instead, pressure varies from location to location and acts normal to each surface element. The resultant is therefore an integral over the complete surface:

$$\vec{F} = -\int_S p \vec{n} dA + \int_S \vec{\tau} dA$$

The first term represents pressure forces and the second represents viscous shear forces. The numerical solver evaluates these contributions over the computational surface. This explains why pressure extrema alone are insufficient to reconstruct the reported 574.656 N and 576.663 N forces.

9.4 Aerodynamic Coefficients

If the free-stream velocity V_∞ , density ρ , reference area S , and the desired force component are known from the CFD setup, the corresponding aerodynamic coefficient can be calculated from:

$$C = F / (\frac{1}{2} \rho V_\infty^2 S)$$

Because the supplied result view does not provide a complete, unambiguous set of those reference quantities, a numerical C_L or C_D is intentionally not fabricated here. This preserves the technical integrity of the report.

10. Discussion

The project demonstrates the complete conceptual chain from aerodynamic profile to engineering result. The NACA 4412 equations define the profile, SOLIDWORKS converts that profile into a three-dimensional wing, and Flow Simulation evaluates the resulting flow field.

The pressure contour confirms a non-uniform surface pressure distribution, while the flow trajectories provide a visual representation of the surrounding fluid motion. The integrated force results quantify the aerodynamic loading predicted by the simulation.

From an engineering perspective, the most important point is that the CAD geometry and CFD solution are directly coupled: changing the airfoil, wing planform, angle of attack, or flow conditions would change the pressure field and consequently the integrated aerodynamic forces.

11. Limitations and Engineering Considerations

- The report uses only CFD quantities that are visible in the supplied project evidence.
- The complete inlet velocity, fluid properties, angle of attack, reference area, and mesh statistics are not reproduced numerically because they are not unambiguously visible in the provided result frame.
- The displayed pressure extrema are not equivalent to an average pressure difference across the wing.
- A single CFD case does not establish a complete lift/drag polar.
- CFD results should ideally be supported by mesh-independence testing and comparison with experimental or published data.

12. Skills and Learning Outcomes

- Applied NACA airfoil mathematics to aerospace geometry.
- Developed practical SOLIDWORKS 3D CAD skills.
- Learned the workflow of external-flow CFD analysis.
- Developed familiarity with computational meshing and post-processing.
- Learned to interpret pressure contours and flow trajectories.
- Connected pressure distribution with integrated aerodynamic force.
- Applied fluid-mechanics equations to a real engineering model.
- Improved technical documentation, engineering judgment, and presentation.

13. Future Work

- Run the wing at multiple angles of attack and construct lift and drag curves.
- Extract C_L and C_D using confirmed reference conditions.
- Perform a mesh-independence study.
- Compare the CFD results with published NACA 4412 experimental data.
- Investigate Reynolds-number effects.
- Study pressure coefficient distribution along selected spanwise stations.
- Evaluate alternative wing sweep, taper, and twist configurations.
- Use CFD loading as an input for structural finite-element analysis.
- Perform aerodynamic optimization for a specified aircraft mission.

14. Applications

- Preliminary aircraft and UAV wing design.
- Concept-level aerodynamic assessment.
- Educational and academic CFD studies.
- Comparison of airfoil and wing configurations.
- Preliminary aerodynamic load estimation.
- Preparation of geometry for advanced CFD and FEA workflows.

15. Conclusion

The CodeAlpha Virtual Internship project successfully integrated aerospace CAD modeling and CFD analysis through the design and evaluation of a NACA 4412 aircraft wing. The project progressed from the mathematical definition of the airfoil to a three-dimensional SOLIDWORKS model and finally to a Flow Simulation analysis.

The supplied CFD results show a displayed pressure range of 100,553.29–101,895.63 Pa, a normal force of 574.656 N, and a resultant force of 576.663 N. The pressure contour and flow trajectories provide a qualitative view of the flow field, while the surface-parameter results provide quantitative aerodynamic loading.

The mathematical development in this report establishes the physical basis for the simulation: conservation of mass describes the flow, Bernoulli's relation provides a simplified pressure–velocity interpretation, and surface integration explains how the distributed pressure and shear stresses become a net aerodynamic force. The project therefore represents a meaningful introductory aerospace engineering workflow and provides a strong foundation for future CFD validation, optimization, and aero-structural analysis.

16. References

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